

Project Initialization and Planning Phase

Date	18 July 2026
Team ID	SWTID-2026-5023
Project Title	Rising Waters (Flood Prediction System)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to improve flood prediction and disaster preparedness using artificial intelligence and real-time environmental data. It addresses the challenges of delayed flood warnings and inaccurate risk assessment, helping authorities and communities respond more effectively. The project enhances public safety, reduces flood-related losses, and supports timely evacuation. Key features include an AI-based flood prediction model, real-time water level monitoring, rainfall analysis, and risk zone visualization.

Project Overview	
Objective	Develop an AI-based flood prediction and early warning system that analyzes rainfall, river water level, and weather data to provide timely flood alerts and reduce disaster risks.
Scope	The system predicts flood risks using machine learning, displays risk levels on a dashboard, and sends real-time alerts to users and disaster management authorities.
Problem Statement	
Description	Existing flood warning systems often provide delayed or inaccurate predictions, making it difficult for people to prepare for flood events.
Impact	Delayed flood warnings result in loss of lives, property damage, disruption of transportation, agriculture losses, and economic impact.
Proposed Solution	
Approach	Build an AI-based prediction model using historical flood, rainfall, river level, and weather datasets to forecast flood risk and generate early alerts.
Key Features	• AI-based flood prediction using Machine Learning • Real-time weather and rainfall monitoring • SMS/Email/App notifications • Risk visualization dashboard • Historical data analysis

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU	Intel i5/Ryzen 5 or above
Memory	RAM specifications	8 GB
Storage	Dataset & Model Storage	256 GB SSD
Software		
Frameworks	ML Framework	Python, Flask/FastAPI
Libraries	ML libraries	scikit-learn, pandas, numpy, matplotlib
Development Environment	IDE	VS Code / PyCharm / Jupyter Notebook
Data		
Dataset	Historical rainfall, river level and flood records	CSV format (Kaggle / Government Flood Dataset)